

Allen Tran

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Education

University of Southern California, Viterbi School of Engineering

Master of Science in Computer Science

Los Angeles, CA

January 2025 – May 2026

University of Southern California, Viterbi and Marshall School of Business

Bachelor of Science in Computer Science and Business Administration

Los Angeles, CA

August 2022 – May 2026

Certificates and Awards: Codepath Intermediate Technical Interview, Dean's List: Fall 2022 – Spring 2025

GPA: 3.65

Relevant Coursework: Data Structures & OOP, Advanced Algorithms, Machine Learning, Information Retrieval

Experience

ServiceNow

Machine Learning Engineer Intern

May 2025 – August 2025

Santa Clara, California

- Engineered scalable infrastructure for evaluating **agentic router prompts** across multiple **LLMs (GPT-4o, Claude, Gemini, LLaMA)**, ensuring high-throughput prompt tuning and reproducible benchmarking.
- Built and maintained a robust **evaluation pipeline** using **Python, PyTorch**, and internal tooling to measure **routing accuracy, tool invocation success, and task completion rates** across tens of thousands of model outputs.
- Designed and integrated **AutoChat** and **Judge LLM** systems to simulate and evaluate full multi-turn conversations, enabling scalable, automated QA for conversational pipelines.
- Collaborated with linguists to deploy **labeled datasets** into production-grade workflows, leveraging orchestration logic and prompt-specific templates for high-fidelity decision benchmarking.
- Merged **6** pull requests and delivered key insights to **agentic-adjacent** teams, highlighting the most effective models for specific agent tasks.

Interaction Lab

Undergraduate Research Assistant

August 2023 – May 2026

Los Angeles, California

- Implement pre-trained **LLMs**(CLIP model) for generating image embeddings from the MuFaSAA Dataset containing data on **165** robots and over **1,200** entries. Identify algorithms, reducing computational time by **87** percent.
- Hyperparameter tuning SVR and transformers to predict social and functional expectations of robots based on design features and metaphors with **Optuna**. Predict **R-squared** values over **0.7** in warmth, discomfort, and competency.
- Collaborate with PhD mentor researching **Vision-Language Models** for predicting explanatory metaphors for robot designs, aiming to enhance user understanding and engagement with robotic technologies.

ServiceNow

Software Engineer Intern

May 2024 – August 2024

Santa Clara, California

- Automated test scripts and story testing in **Java** using **Selenium** for regression and coverage testing across ServiceNow's internal tools – **lists, forms, and streams**. Merged **5** pull requests into the main branch.
- Engineered an **AI-driven** extension that auto-fills build requests and summarizes PRs. Built using **TypeScript, Node, GitHub and OpenAI API**. Streamlines code merging process within ServiceNow, reducing time by **5** seconds.
- Deployed a **Mattermost** bot for live updates on defects and stories. Built on **Javascript, JSON**, and **Jenkins'** pipeline.

CodePath

Tech Fellow (Advanced Interview Prep)

May 2024 – August 2024

Remote

- Facilitated student understanding of **data structures and algorithms** as a TA for the **TIP 103** cohort.
- Applied structured teaching strategies such as **UMPIRE** and **cold-calling** to improve interview skills.
- Mentored groups of **6 students** at a time, fostering a collaborative and inclusive environment to support ideation and problem-solving.

Projects

Adaptive RAG Research Paper | *Python, Pytorch*

[link to paper](#) | August 2025 – May 2025

- Co-author a research paper proposing a novel RAG pipeline that selectively triggers retrieval based on an LLM's **uncertainty estimation**, reducing token usage and improving efficiency.
- Develop an inference-time controller that computes entropy-based thresholds to determine when external context is required vs. when parametric knowledge suffices.
- Benchmark across QA datasets (**SQuAD, Natural Questions, TriviaQA**) with evaluation on accuracy, latency, and token savings compared to standard RAG baselines.

BizLang | *Javascript, React.JS, Flask*

trybizlang.com | August 2023 – Present

- Built an app like **Duolingo** that develops one's industry language through interactive learning.
- Incorporated **gamified** elements such as **Wordle, news feeds** from GNews API, and **flashcards** to chat with to enhance user engagement. Backed with over **1000** jargons to learn from.
- Attracted over **100** users and actively conducting beta testing with **30** healthcare professionals in **Ethiopia**.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, Java, TypeScript, PHP, R

Frameworks/Tools: React, Flask, Django, PyTorch, TensorFlow, Jenkins, Docker, GCP, AWS, Git

Libraries: Hugging Face, Pandas, NumPy, Scikit-learn, Firebase

Core Skills: ML Ops, Prompt Engineering, NLP, Data Analysis, Distributed Systems, CI/CD, REST APIs